

Computer Vision

Compiled by:
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Textbook or References

Textbook:

- Rick Szeliski, Computer Vision: Algorithms and Applications online at: <http://szeliski.org/Book/>
- Together with the book, other resources derived or related to the above textbook will be used.

What is Computer Vision?

Computer Vision is the field of AI that enables computers to:

- perceive,
- process,
- analyze,
- and understand visual information from images and videos.

Computer Vision - Key Objective

- Transform raw pixels into meaningful understanding.
- Goal of computer vision: perceive the “story” behind the picture
- Compute properties of the world
 - 3D shape
 - Names of people or objects



0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

Human Vision vs Computer Vision

Human Vision	Computer Vision
Biological System	Computational System
Uses Eyes + Brain	Uses cameras + algorithms
Learns Naturally	Learns from Data
Context-Aware	Requires Explicit Modeling

Can computers match human perception?

- Yes and no
 - humans are better at “hard” things, are more robust, and make inferences quickly and cheaply
- But huge progress
 - Accelerating in the last 10 years due to deep learning, large vision-language models

What is considered “hard” keeps changing





What do you see in this picture? What do you think happened before this picture was taken?



In the picture, we see the famous Montparnasse derailment from 1895. It shows a steam locomotive that has crashed through the station wall and is hanging precariously over the street below. The photograph captures the aftermath of a dramatic accident where the train failed to stop while approaching the buffer stop at the Gare Montparnasse terminal in Paris.

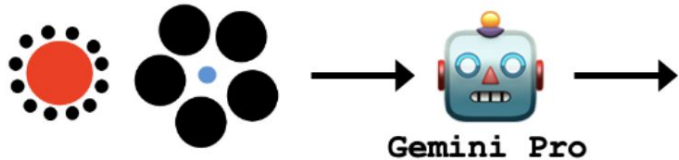
Before this picture was taken, the train, arriving from Granville to Paris, was running late. The driver, trying to make up for lost time, was likely going too fast and unable to stop the train in time upon reaching the station. The train overran the buffer stop, crossed the station concourse, crashed through a 60 cm thick wall, and ended up nosediving onto the Place de Rennes below. Remarkably, only one person, a woman selling newspapers on the street, was killed by falling masonry, and several others were injured. This image has become a symbol of dramatic accidents and is frequently used to illustrate the need for careful management of technology and speed.

Current models still make very silly mistakes

Prompt: "Which is bigger, the blue circle or the red circle?"

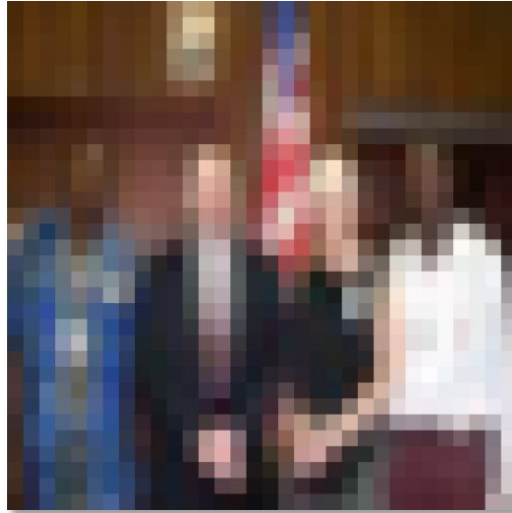
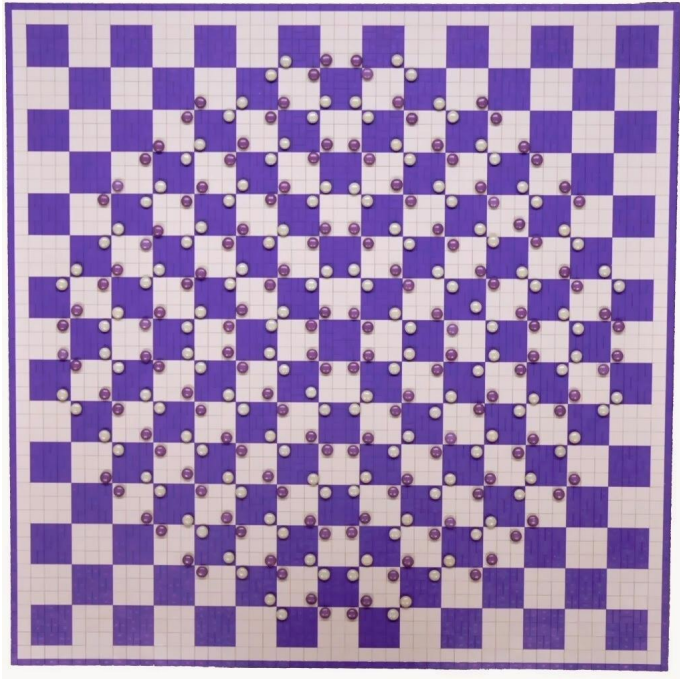


"This is a classic optical illusion! The red and blue circles are actually the same size. Our brains tend to perceive the red circle as smaller because it's surrounded by larger black circles. The blue circle, surrounded by smaller black circles, appears larger in comparison."



"This is a classic optical illusion! The red and blue circles are actually the same size. Our brains tend to perceive the red circle as larger because it's surrounded by smaller circles, while the blue circle is surrounded by larger circles. This is an example of the Ebbinghaus illusion."

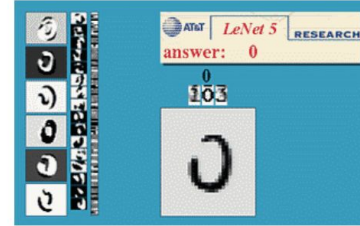
Human Perception - Shortcomings vs strength



Computer Vision Applications

Industrial Applications

- Healthcare
- Agriculture
- Surveillance
- Manufacturing
- Robotics
- Retail
- Autonomous systems



(a)



(b)



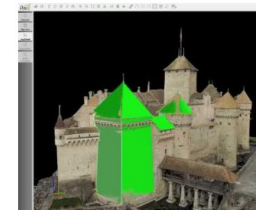
(c)



(d)



(e)



(f)

Computer Vision Applications

Consumer Level Applications

- Stitching
- Exposure bracketing
- Morphing
- 3D modeling
- Video match move and stabilization
- Photo-based walkthroughs
- Face detection
- Visual authentication



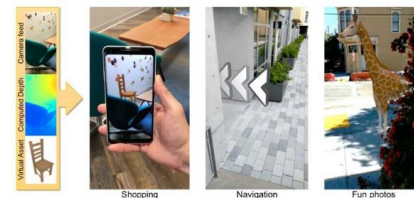
(a)



(b)



(c)



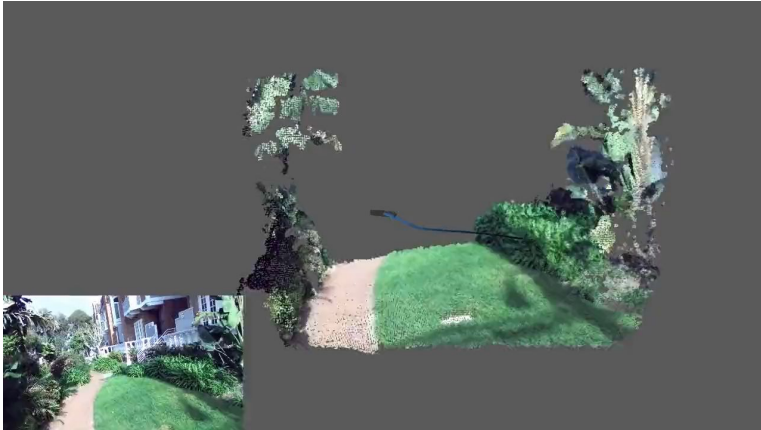
(d)

Goal of Computer Vision

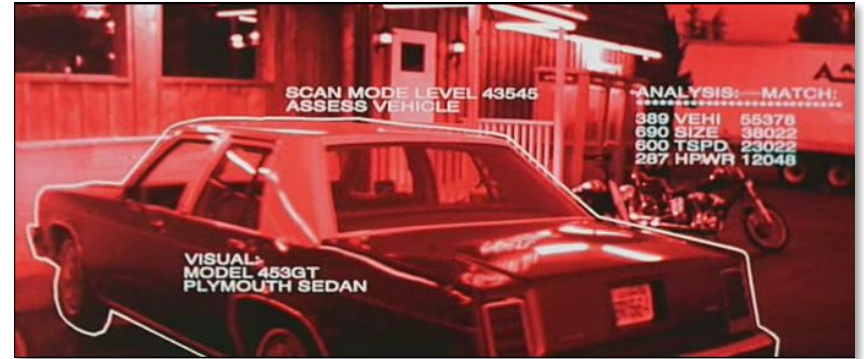
Compute the 3D shape of the world



ZED 2i Camera



Recognize objects and people

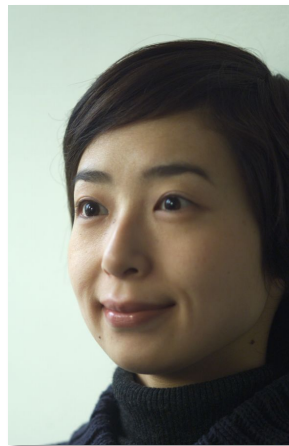


Recognize objects and people



slide credit: Fei-Fei, Fergus & Torralba

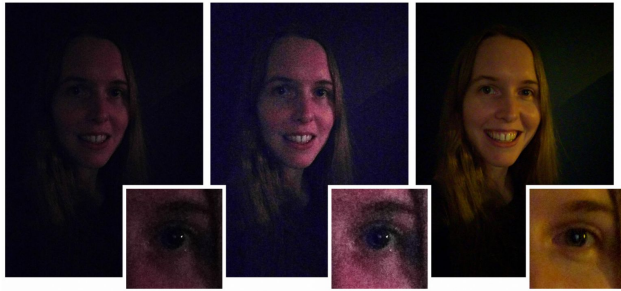
Goal of Computer Vision - Forensics



Goal of Computer Vision - Improve Photos



Super-resolution (source: 2d3)



Low-light photography
(credit: [Hasinoff et al., SIGGRAPH ASIA 2016](#))



Depth of field on cell phone camera
(source: [Google Research Blog](#))



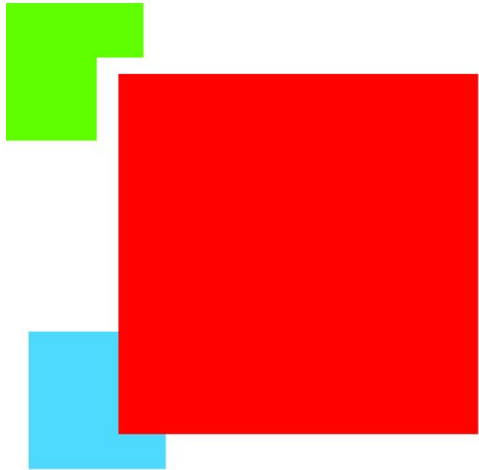
Removing objects
([Google Magic Eraser](#))

Why study Computer Vision

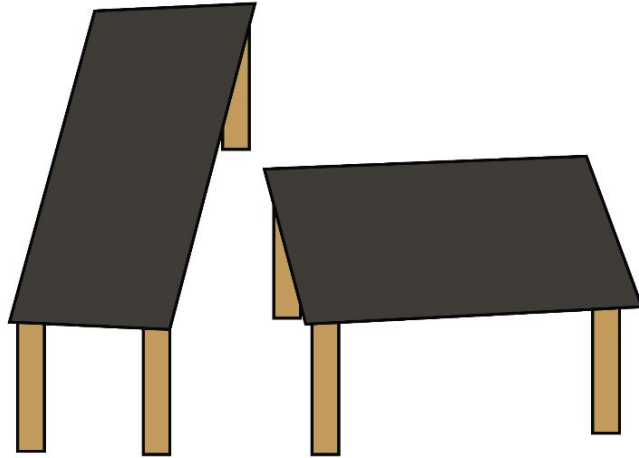
- Billions of images/videos captured per day
- Huge number of potential applications

Why is vision hard?

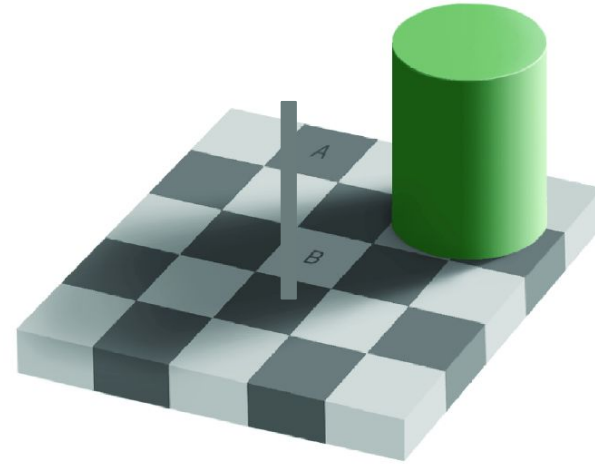
Perception vs. measurement



(a)



(b)



(c)

Why is vision hard?



Viewpoints

Scale



Illumination

Why is vision hard?



Intra-class
variation



Background
clutter



Motion (Source: S.
Lazebnik)



Occlusion

The state of Computer Vision and AI: we are really, really far away.- Andrej Karpathy,2012



Syllabus Discussion (Tentative)